

- 16** The molecules of an ideal gas at thermodynamic temperature T have a root-mean-square speed c .

The gas is heated to temperature $2T$.

What is the new root-mean-square speed of the molecules?

A $\sqrt{2}c$

B $2\sqrt{2}c$

C $2c$

D $4c$

[Turn over

Ans A

$$\frac{1}{2}m(c)^2 = \frac{3}{2}kT$$

$$c \propto \sqrt{T}$$

$$\frac{c_{new}}{c} = \sqrt{\frac{2T}{T}}$$

$$c_{new} = \sqrt{2}c$$

17. Which one of the following statements about the electric potential at a point is correct?